

ADITYA RAJ – CV
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Research Overview

My research explores complex, high-dimensional systems that exhibit collective dynamics and emergent behavior ubiquitous in nature. Using computational and theoretical approaches, I investigate how interactions, coupling structures, and network organization influence system behavior. By combining tools from dynamical systems theory, such as Lyapunov analysis and covariant Lyapunov vectors, with models of spatially extended and networked systems, I aim to build a physical understanding of the spatiotemporal dynamics and pattern formation in complex systems.

Education

1. PhD, Department of Mechanical Engineering, Virginia Tech, Blacksburg, USA, *Aug 2021 – Exp Dec 2026*
2. M. Tech., Department of Mechanical Engineering, IIT Dhanbad, India, *Aug 2017 – May 2019*
3. B.Tech., Mechanical Engineering, MIT Muzaffarpur, India, *Aug 2011 – May 2015*

Summer program:

4. Complex Systems Summer School 2025, Santa Fe Institute, Santa Fe, New Mexico, USA. *June, 2025*

Work Experience

Graduate Teaching Assistant

Department of Mechanical Engineering, Virginia Tech, Blacksburg, USA – (*Spring, 2025, Fall 2025 -*)
GTA in the Engineering Analysis Using Numerical Methods course

- Advised students on assignments and quizzes
- Graded recitation quizzes and exams
- Conducted recitations

Research and Development Intern

Aegis Cooling Technologies, Greenville, USA – (*June 2026 - Aug 2026*)

- Executed steady-state and transient thermal performance testing of single-phase and two-phase cold plates and a liquid-cooled coolant distribution unit (CDU), validating chip-temperature reliability across the operating envelope.
- Conducted vibration testing and frequency-domain vibration analysis on the flow manifold to map flow regimes and identify flow-induced excitation signatures across operating conditions.
- Operated, calibrated, and maintained laboratory test rigs — instrumenting sensors, DAQ, and control hardware, and performing refrigerant charging/discharging under established safety protocols — to ensure measurement accuracy and run-to-run repeatability.
- Monitored experiments in real time and troubleshoot abnormal thermal and flow behavior; performed data reduction, statistical analysis, and uncertainty evaluation on validated experimental datasets.
- Authored the Installation, Operation, and Maintenance (IOM) manual for the CDU, and documented test procedures, configurations, and results; communicated findings through technical reports, plots, and presentations.

Graduate Teaching Assistant

Department of Mechanical Engineering, Virginia Tech, Blacksburg, USA – (*Spring, 2025, Fall 2025 -*)
Lead GTA in Fluid Dynamics Lab.

- Conducted laboratory sessions
- Graded student reports
- Advised students in office hours

Graduate Research Assistant

Paul Research Group – Nonlinear Dynamics and Pattern Formation, Department of Mechanical Engineering, Virginia Tech, Blacksburg, USA – (*May 2022 - Dec 2024, Summer 2025*)

- Exploring spatiotemporal chaos in spatially extended systems influenced by short- and long-range couplings, as seen in fluid dynamics equations such as the Navier-Stokes equations.

- Investigating coupled map lattices (CMLs) and partial differential equations (PDEs), utilizing tools such as covariant Lyapunov vectors (CLVs) to analyze chaotic behavior in high-dimensional systems.
- Studying pattern formation in Rayleigh–Bénard convection using the Generalized Swift-Hohenberg equation (GSH)- and ongoing investigation that explores how mean flow strength can affect the tangent-space dynamics and spiral defect chaos.
- Developed numerical solvers for the numerical solutions and tangent-space calculations of CMLs and GSH equation. Used finite difference method, pseudospectral method, various Runge-Kutta solvers, and exponential time integration for PDEs.

Graduate Research Assistant

CEHMS Group, Department of Mechanical Engineering, Virginia Tech, Blacksburg, USA – (*Jan 2022 - May 2022*)

- Studied bistable energy harvesting, especially using the phenomenon of stochastic resonance.
- Contributed in the construction of bistable energy harvester capable of snap-through mechanism.
- Assisted in performing the experiments to characterize the stiffness of bio-inspired bistable energy harvesting.

Graduate Teaching Assistant

Department of Mechanical Engineering, Virginia Tech, Blacksburg, USA – (*Aug 2021 - Dec 2021*)
GTA in ME 3624 Mechanical Design Lab.

- Conducted laboratory sessions
- Graded student reports
- Advised students in office hours

Project Assistant III

Aerosystems Laboratory, CSIR-CMERI, Durgapur, India – (*Aug 2019 - Mar 2020*)

Project: Design of Pressure Regulating and Shut-Off Valve (PRSOV) for the Environmental Control System (ECS) of aircraft.

- Drafting and Solid Modeling.
- Performed inspection of blueprints.
- Brainstormed for design improvements.
- Contributed to building mathematical models for flows through orifices.

Teaching Assistant

Department of Mechanical Engineering, IIT(ISM), Dhanbad, India – (*Aug 2018 - May 2019*)

- Taught students to solve quiz problems.
- Proctored quiz sessions.

Publications

1. **A. Raj** and M. R. Paul. Spatiotemporal Chaos with Extended Spatial Interactions. *International Journal of Bifurcation and Chaos* (2026): 2750010.
2. **A. Raj** and M. R. Paul. Exploring the role of diffusive coupling in spatiotemporal chaos. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 34(10), 2024. <https://doi.org/10.1063/5.0210661>
3. **A. Raj**, J. P. Varun, and P. K. Mahato. Fabrication and vibration damping analysis of basalt fiber reinforced composite beam. In AIP Conference Proceedings, volume 2134. AIP Publishing, 2019.

Presentations

1. **A. Raj** and M. R. Paul, A dynamical systems perspective on spatial couplings in large chaotic systems. Presentation at Fall Fluid Mechanics Symposium 2025, Virginia Tech.
2. **A. Raj** and M. R. Paul, Building a physical understanding of the role of spatial couplings in high-dimensional chaotic systems. Poster presentation at Walter O'Brien Graduate Research Symposium 2025, Virginia Tech.
3. **A. Raj** and M. R. Paul, Exploring the role of long-range coupling on chaotic fluid flows using Lyapunov vectors, Bulletin of the American Physical Society (2024). Presentation at APS DFD 2024.
4. **A. Raj** and M. R. Paul, Exploring the role of long-range coupling on chaotic fluid flows using Lyapunov vectors). Presentation at Fall Fluid Mechanics Symposium 2024.
5. **A. Raj** and M. R. Paul, Using covariant Lyapunov vectors to explore spatiotemporal chaos: from coupled maps to PDEs. Poster presentation at Walter O'Brien Graduate Research Symposium at Virginia Tech (2024).

6. **A. Raj** and M. R. Paul, Using Covariant Lyapunov Vectors to Investigate the Role of Spatial Interactions in Chaotic Fluid Systems, Bulletin of the American Physical Society (2023). Presentation at APS DFD 2023.
7. **A. Raj** and M. R. Paul, The Spatiotemporal Chaos of Coupled Maps: Insights from the Covariant Lyapunov Vectors. Presentation at SIAM DS 23 (2023).
8. **A. Raj** and M. R. Paul, Building a Physical Understanding of Spatiotemporal Chaos using Covariant Lyapunov Vectors. Presentation at SIAM SEAS (2023).
9. **A. Raj** and M. R. Paul, Exploring the Spatiotemporal Chaos of Lattices of Coupled Maps with Diffusive and Convective Spatial Interactions. Poster Presentation in Walter O'Brien Graduate Research Symposium at Virginia Tech (2023).
10. **A. Raj** and M. R. Paul, Exploring the Role of Spatial Coupling in Spatiotemporal Chaos Using Covariant Lyapunov Vectors, Bulletin of the American Physical Society (2023). Presentation at APS March Meeting 2023.
11. **A. Raj** and M. R. Paul, Using Covariant Lyapunov Vectors to Explore Chaotic Dynamics with Long-Range Spatial Coupling, Bulletin of the American Physical Society (2022). Poster Presentation at APS DFD 2022.

Awards

1. Complex Systems Summer School Scholarship, Santa Fe Institute, New Mexico, USA (Partial funding). (*June 2025*)
2. Graduate Aptitude Test in Engineering (GATE) Scholarship, Ministry of Human Resource Development, Government of India. (*July 2017 - May 2019*)

Skills

- MATLAB: With 10+ years of experience, I have developed code for the numerical computations of coupled map lattices, partial differential equations and small projects involving computational fluid dynamics such as lid-driven cavity. I have also used it for frequency domain analysis of vibrations of composite beams.
- Python: Intermediate level proficiency.
- Academic writing: Authored two journal papers and a conference paper.
- Autodesk Inventor: Used this 3D modeling tool proficiently at my stint at CSIR-CMERI Durgapur.
- ANSYS: Used it for a term project involving the calculations of a fluid flow in a lid-driven cavity.

Extra-curricular

- Executive Board member of WeMET (Welcome to Mechanical Engineering) at Virginia Tech, an initiative that promotes wellness among graduate students in ME. (*Aug 2025 -*)
- System administrator for the Fall Fluid Mechanics Symposium 2024 at Virginia Tech.
 - Built the website for the symposium and organized the sessions.
 - Chaired a session in the symposium.
- Outreach chairperson for the graduate student organization MEGSC (Mechanical Engineering Graduate Student Council) at Virginia Tech. (*March 2023 - July 2025*)
 - Designed and managed the website and the SharePoint site for MEGSC.
- Peer mentoring
 - Peer mentored for ME department's Graduate Peer Mentor Program at Virginia Tech (*Fall 2024 semester*).
 - Peer mentored for MEGSC's peer mentoring program (Fall 2023 semester).
- Served as the president of the graduate student organization GEA (Graduate Engineering Alliance) at Virginia Tech (Jan 2023 - Dec 2023).
- Ranked 1st in a poster drawing competition in undergrad (BTech) at MIT Muzaffarpur.

Hobbies and interests

Chess, table-tennis, hiking, movies, art.